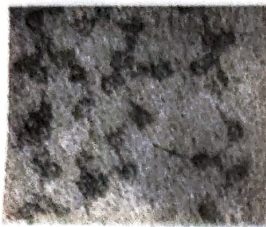


1. Identify the given images below in a and b, and write a short description. 3

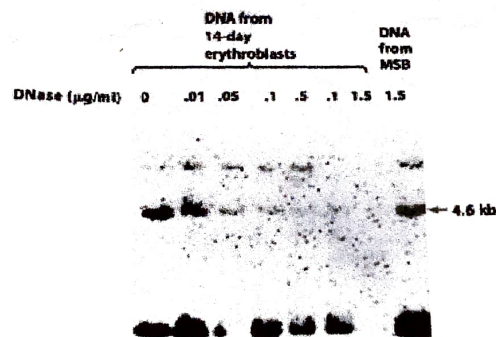


(a)



(b)

2. Describe with the help of a diagram how the mechanism of alternative lengthening of telomeres (ALT) differs from the normal telomere extension pathway in yeast cells. 3
3. The loop extrusion hypothesis states that the genomic DNA loop sizes vary depending on the resident time of Cohesin. How would you demonstrate this experimentally? 3
4. Nuclei from 14-day-old erythroblast and MSB cells of chicken were treated with DNase I. The extracted DNA was digested with *Bam*HI. Explain why the experimenter noted two distinct digestion patterns. 3



5. For centromere specification, two conditions need to be met by the chromosome. Write the experimental observations that support this hypothesis. 3